

RESEARCH SUMMARY

My research focuses on **auditing the security and privacy of machine learning (ML) systems**, particularly large language models (LLMs). My work makes AI risks measurable across the AI lifecycle by (i) auditing **training-data leakage** through membership inference; (ii) tracing **data and model provenance** across ML development pipelines; and (iii) identifying and mitigating privacy risks arising from **LLM misuse**. I also develop data synthesis algorithms to protect sensitive information based on **differential privacy**.

EDUCATION

Purdue University Ph.D. in Computer Science (Advisor: Ninghui Li)	2023 – 2027 (Expected) USA
Zhejiang University M.E. in Computer Technology. <i>Rank 1/49.</i>	2020 – 2023 China
East China Normal University B.E. in Data Science. <i>GPA 3.61/4 (rank 1/23).</i>	2016 – 2020 China

SELECTED HONORS AND AWARDS

National Winner, Innovation Bowl (Purdue Team Lead)	2026
NDSS Fellowship , Internet Society (1 of 24 worldwide)	2026
Ross Fellowship , Purdue University	2023
Herbold Scholarship , Purdue University (1 of 7)	2023
Presidential Doctoral Excellence Award , Purdue University (1 of 150)	2023
Excellent Master's Dissertation , China (1 of 43)	2023
Provincial Outstanding Graduate , Zhejiang, China	2023
National Scholarship , China (0.1%)	2022
National Scholarship , China (0.1%)	2021

WORK EXPERIENCE

Research Intern, Perplexity AI Mentor: Dr. Ninghui Li Characterize privacy and safety risks of agents in real-world settings, design practical security guardrails for LLM agents, and establish a defense-in-depth architecture deployed in Perplexity's products.	2026.05 – Present
Research Intern, Alibaba Qwen Mentor: Dr. Bolin Ding Identified a new privacy threat to online pseudonymity arising from LLM misuse, and developed the <i>first</i> agentic framework demonstrating the feasibility of automated deanonymization with LLMs. Results published in ACL 2026 and the IEEE Data Engineering Bulletin .	2024.05 – 08
Research Intern, Microsoft Research Asia Mentor: Dr. Xing Xie Designed a new explainable recommendation algorithm that combines the interpretability of sparse methods with the expressiveness of dense approaches. The algorithm was deployed on Xbox Gaming, with results published at WWW 2023 . The project also resulted in a U.S. patent and received the Excellent Internship Award (top 10%).	2022 – 2023

PUBLICATIONS

First/Co-First	11 publications at CCS (×2), USENIX Security (×2), NDSS, ACL, VLDB, WWW, SIGIR, TKDE, and IEEE Data Eng. Bulletin.
Mentoring	Students I mentored published 4 papers at USENIX Security, SIGIR, and ICDE (×2).

*equal contribution, †mentored student

CCS 2026	Privacy Leakage from a Thousand Words: Sub-Pixel Location Recovery from Dot Maps. Yuntao Du [*] , Tanishq Praveen Pauskar [†] , Hao Wang, Jing Su, Ninghui Li. <i>In ACM Conference on Computer and Communications Security (CCS), 2026.</i>
ACL 2026	Automated Profile Inference with Language Model Agents. Yuntao Du, Zitao Li, Bolin Ding, Yaliang Li, Hanshen Xiao, Jingren Zhou, Ninghui Li. <i>In Findings of the Association for Computational Linguistics (ACL), 2026.</i>
USENIX 2026	Imitative Membership Inference Attack. Yuntao Du, Yuetian Chen, Hanshen Xiao, Bruno Ribeiro, Ninghui Li. <i>In USENIX Security Symposium (USENIX Security), 2026.</i>
USENIX 2026	Membership Inference Attacks on Tokenizers of Large Language Models. Meng Tong [†] , Yuntao Du [*] , Kejiang Chen, Weiming Zhang, Ninghui Li. <i>In USENIX Security Symposium (USENIX Security), 2026.</i>
USENIX 2026	Window-based Membership Inference Attacks Against Fine-tuned Large Language Models. Yuetian Chen, Yuntao Du, Kaiyuan Zhang, Ashish Kundu, Charles Fleming, Bruno Ribeiro, Ninghui Li. <i>In USENIX Security Symposium (USENIX Security), 2026.</i>
ICLR 2026	Membership Inference Attacks Against Fine-tuned Diffusion Language Models. Yuetian Chen, Kaiyuan Zhang, Yuntao Du, Edoardo Stoppa, Charles Fleming, Ashish Kundu, Bruno Ribeiro, Ninghui Li. <i>In International Conference on Learning Representations (ICLR), 2026.</i>
NDSS 2026	Cascading and Proxy Membership Inference Attacks. Yuntao Du, Jiacheng Li, Yuetian Chen, Kaiyuan Zhang, Zhizhen Yuan, Hanshen Xiao, Bruno Ribeiro, Ninghui Li. <i>In Network and Distributed System Security Symposium (NDSS), 2026.</i>
IEEE Data Eng. Bulletin	Beyond Data Privacy: New Privacy Risks for Large Language Models. Yuntao Du, Zitao Li, Ninghui Li, Bolin Ding. <i>In IEEE Computer Society Data Engineering Bulletin. 2025.</i>
CCS 2025	Systematic Assessment of Tabular Data Synthesis. Yuntao Du, Ninghui Li. <i>In ACM Conference on Computer and Communications Security (CCS), 2025.</i>
USENIX 2025	SOFT: Selective Data Obfuscation for Protecting LLM Fine-tuning against Membership Inference Attacks. Kaiyuan Zhang, Siyuan Cheng, Hanxi Guo, Yuetian Chen, Zian Su, Shengwei An, Yuntao Du, Charles Fleming, Ashish Kundu, Xiangyu Zhang, Ninghui Li. <i>In USENIX Security Symposium (USENIX Security), 2025.</i>
Before Ph.D.	
ICDE 2024	Real-Time Trajectory Synthesis with Local Differential Privacy. Yujia Hu [†] , Yuntao Du, Zhikun Zhang, Ziquan Fang, Lu Chen, Kai Zheng, Yunjun Gao. <i>In IEEE International Conference on Data Engineering (ICDE), 2024.</i>
VLDB 2023	LDPTTrace: Locally Differentially Private Trajectory Synthesis. Yuntao Du, Yujia Hu [†] , Zhikun Zhang, Ziquan Fang, Lu Chen, Baihua Zheng, Yunjun Gao. <i>In International Conference on Very Large Data Bases (VLDB), 2023.</i>

WWW 2023	Towards Explainable Collaborative Filtering with Taste Clusters Learning. Yuntao Du, Jianxun Lian, Jing Yao, Xiting Wang, Mingqi Wu, Lu Chen, Yunjun Gao, Xing Xie. <i>In ACM Web Conference (WWW), 2023.</i>
SIGIR 2023	Knowledge-refined Denoising Network for Robust Recommendation. Xinjun Zhu[†], Yuntao Du, Yuren Mao, Lu Chen, Yujia Hu, Yunjun Gao. <i>In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2023.</i>
ICDE 2023	FLBooster: A Unified and Efficient Platform for Federated Learning Acceleration. Zhihao Zeng[†], Yuntao Du, Ziquan Fang, Lu Chen, Shiliang Pu, Guodong Chen, Hui Wang, Yunjun Gao. <i>In IEEE International Conference on Data Engineering (ICDE), 2023.</i>
TKDE 2023	Finding Materialized Models for Model Reuse. Minjun Zhao, Lu Chen, Keyu Yang, Yuntao Du, Yunjun Gao. <i>In IEEE Transactions on Knowledge and Data Engineering (TKDE), 2023.</i>
TKDE 2023	MetaKG: Meta-learning on Knowledge Graph for Cold-start Recommendation. Yuntao Du, Xinjun Zhu[†], Lu Chen, Ziquan Fang, Yunjun Gao. <i>In IEEE Transactions on Knowledge and Data Engineering (TKDE), 2023.</i>
KDD 2022	Spatio-Temporal Trajectory Similarity Learning in Road Networks. Ziquan Fang, Yuntao Du, Xinjun Zhu, Danlei Hu, Lu Chen, Yunjun Gao, Christian S. Jensen. <i>In ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2022.</i>
SIGIR 2022	Self-Guided Learning to Denoise for Robust Recommendation. Yunjun Gao, Yuntao Du, Yujia Hu[†], Lu Chen, Xinjun Zhu, Baihua Zheng. <i>In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2022.</i>
SIGIR 2022	HAKG: Hierarchy-Aware Knowledge Gated Network for Recommendation. Yuntao Du, Xinjun Zhu[†], Lu Chen, Baihua Zheng, Yunjun Gao. <i>In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2022.</i>
ICDE 2021	E²DTC: An End to End Deep Trajectory Clustering Framework via Self-Training. Ziquan Fang, Yuntao Du, Lu Chen, Yujia Hu, Yunjun Gao, Gang Chen. <i>In IEEE International Conference on Data Engineering (ICDE), 2021.</i>
VLDB 2021	MDTP: A Multi-source Deep Traffic Prediction Framework over Spatio-Temporal Trajectory Data. Ziquan Fang, Lu Pan, Lu Chen, Yuntao Du, Yunjun Gao. <i>In International Conference on Very Large Data Bases (VLDB), 2021.</i>

PATENTS

U.S. Patent	Explainable Cluster-based Collaborative Filtering. Mingqi Wu, Jianxun Lian, Yuntao Du, Jing Yao, Xiting Wang, Bei Lu, Xing Xie. <i>U.S. Patent No. 12,524,502 B2, 2026.</i>
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TEACHING

Instructor, CS 556: Data Security and Privacy, Purdue University.

Fall 2026

Teaching Assistant, CS 556: Data Security and Privacy, Purdue University.

Spring 2025

MENTORING

Edward Chen, high school student, Carmel High School.	2026 – Present
Kimaya Deshpande, B.S. Computer Science, Purdue University.	2026 – Present
Annabelle Nakamura DeFosse, B.S. Computer Engineering, Purdue University.	2026 – Present
Patrick Li, B.S. Computer Science, Stanford University. <i>Publications: 1 manuscript under submission</i>	2025 – Present
Tanishq Praveen Pauskar, B.S. Computer Science, Purdue University. <i>Publications: CCS 2026</i>	2025 – Present
Meng Tong, M.S. Computer Science, University of Science and Technology of China. <i>Publications: USENIX Security 2026</i>	2025 – Present
Hillary Yang, high school student, Carmel High School. <i>Publications: 1 manuscript at arXiv</i>	2024 – 2025
Yujia Hu, M.S. Computer Science, Zhejiang University. <i>Publications: SIGIR 2022, VLDB 2023, ICDE 2024</i>	2022 – 2024
Zhihao Zeng, M.S. Computer Science, Zhejiang University. <i>Publications: ICDE 2023</i>	2022 – 2023
Xinjun Zhu, M.S. Computer Science, Zhejiang University. <i>Publications: SIGIR 2022, SIGIR 2023, TKDE 2023</i>	2022 – 2023

ACADEMIC SERVICE

CONFERENCE REVIEWING

PC, USENIX Security Symposium	2027
PC, International Conference on Very Large Data Bases (VLDB)	2027
PC, ACM Asia Conference on Computer and Communications Security (AsiaCCS)	2027
PC, Neural Information Processing Systems (NeurIPS)	2026
Poster Committee, IEEE Symposium on Security and Privacy (S&P)	2026
PC, International Conference on Learning Representations (ICLR)	2025 – 2026
PC, International Conference on Artificial Intelligence and Statistics (AISTATS)	2025 – 2026
PC, ACM Web Conference (WWW)	2026
PC, ACM Conference on Data and Application Security and Privacy (CODASPY)	2026
PC, International Conference on Web Search and Data Mining (WSDM)	2026
PC, ACM Conference on Information and Knowledge Management (CIKM)	2024 – 2026
PC, ACM Conference on Research and Development in Information Retrieval (SIGIR)	2023 – 2026
PC, International Joint Conference on Neural Networks (IJCNN)	2026 – 2027
PC, Table Representation Learning Workshop (TRL)	2024 – 2025
PC, AAAI Conference on Artificial Intelligence (AAAI)	2023 – 2027
PC, ACM Conference on Information Retrieval in the Asia Pacific (SIGIR-AP)	2023 – 2024

JOURNAL REVIEWING

Reviewer, ACM Computing Surveys (CSUR)
Reviewer, IEEE Transactions on Dependable and Secure Computing (TDSC)
Reviewer, ACM Transactions on Privacy and Security (TOPS)
Reviewer, International Journal on Very Large Data Bases (VLDBJ)
Reviewer, ACM Transactions on Recommender Systems (TORS)
Reviewer, IEEE Transactions on Knowledge and Data Engineering (TKDE)
Reviewer, IEEE Transactions on Big Data (TBD)

INVITED TALKS

CS 4501, Data Privacy , <i>Guest lecture</i> , University of Virginia Membership Inference Attacks: Progress and Directions	2026 Apr
CS 593, Trustworthy Machine Learning , <i>Guest lecture</i> , Purdue University Membership Inference Attacks: History, Recent Progress, and Future Directions	2026 Mar
Research Trend Automated Profile Inference with Language Model Agents	2025 Jun
Graduate Research Symposium , Purdue University Automated Profile Inference with Language Model Agents	2025 Mar
CS 556, Data Security and Privacy , <i>Guest lecture</i> , Purdue University Differentially Private Data Synthesis	2025 Mar
AI Time Towards Explainable Collaborative Filtering with Taste Clusters Learning	2023 Nov
Microsoft Research Asia Explainable Clustering and Cluster-based Collaborative Filtering	2023 Jan